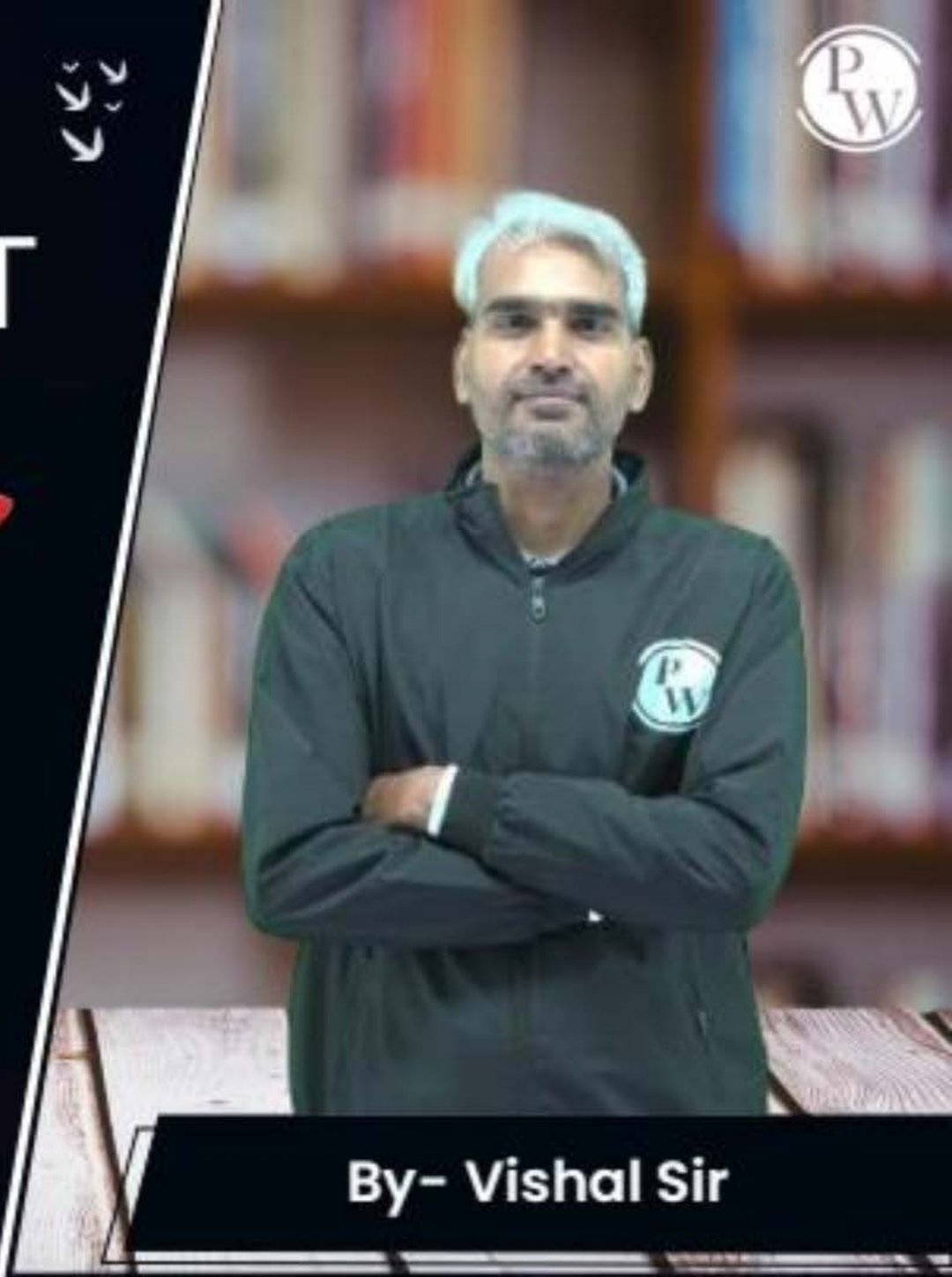
Computer Science & IT



Database Management System

Query Languages

Lecture No. 04



By- Vishal Sir

Recap of Previous Lecture



Topic Practice questions on relational algebra



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Topics to be Covered



- Topic: SQL
- Topic: SQL commands
- Topic: SQL clauses
- Topic: Aggregate functions

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Consider the following relational tables:

Supplier (Sid, Sname, Rating)

Parts (Pid, Pname, Color)

Catalog (Sid, Pid, Cost)

Catalog

Sid	Pid	Cost
S_1	P_1	20
S_1	P_2	30
S_2	P_2	30
S_3	P_3	10

S_1
S_2

S_1	P_2	30
S_2	P_2	30

tuples which are not w.r.t. Max Cost

#Q. Retrieve Sid of the suppliers who have supplied most expensive parts.

$$\pi_{Sid}\left[\text{Catalog} - \pi_{C_1.Sid, C_1.Pid, C_1.Cost}\left(\sigma_{C_1.Cost < C_2.Cost}(C_1 \times C_2)\right)\right]$$

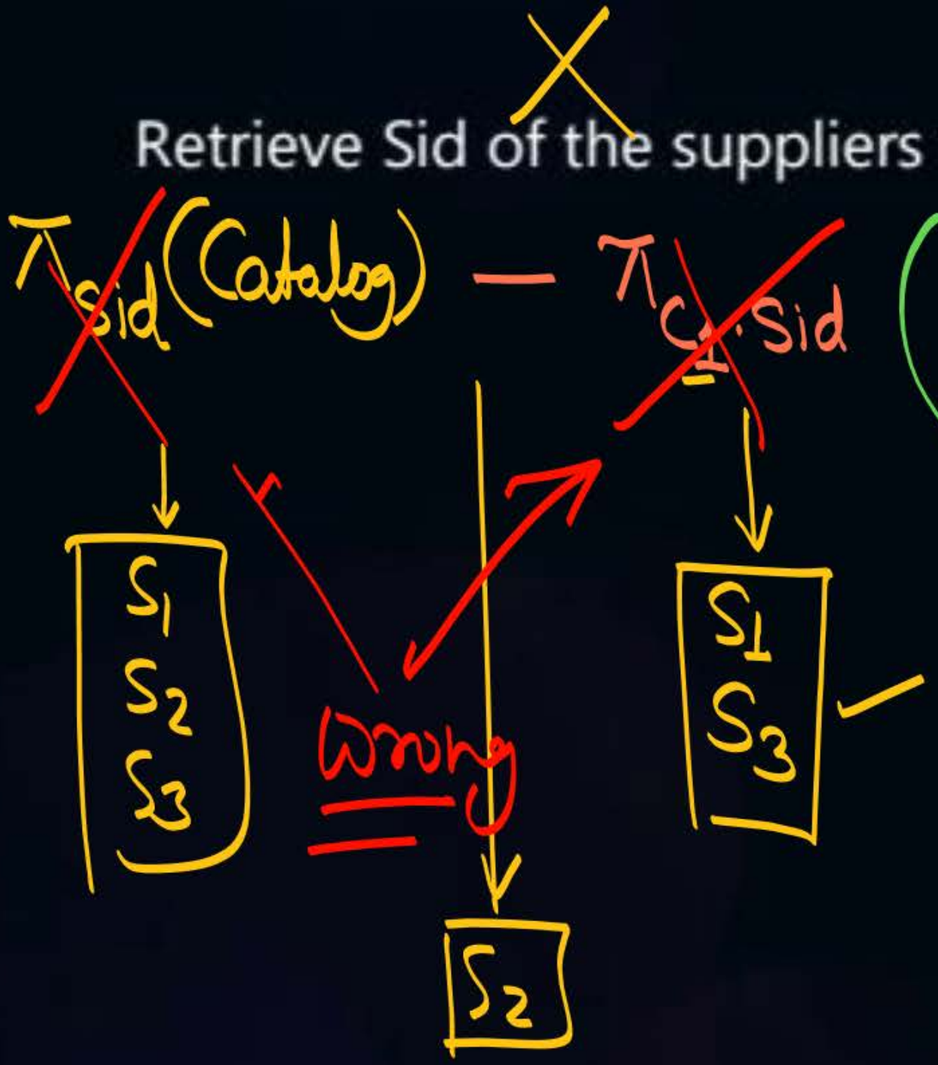
- All tuples of Catalog
- Tuples from Catalog w.r.t Maximum Cost

Sid	Pid	Cost
S_1	P_1	20
S_3	P_3	10

We got the tuples from the Catalog table, in which the Cost is not maximum

Sid	Pid	Cost	Sid	Pid	Cost
S_1	P_1	20	S_1	P_2	30
S_1	P_1	20	S_2	P_2	30
S_1	P_2	30	S_1	P_2	30
S_1	P_2	30	S_2	P_2	30
S_2	P_2	30	S_1	P_2	30
S_2	P_2	30	S_2	P_2	30
S_3	P_3	10	S_1	P_2	30
S_3	P_3	10	S_2	P_2	30

#Q. Retrieve Sid of the suppliers who have supplied most expensive parts.



Sid	Pid	Cost	Sid	Pid	Cost
S_1	P_1	200	S_1	P_1	200
S_1	P_2	200	S_1	P_2	200
S_1	P_3	300	S_1	P_3	300
S_1	P_4	400	S_1	P_4	400
S_2	P_1	200	S_1	P_1	200
S_2	P_2	200	S_1	P_2	200
S_2	P_3	300	S_1	P_3	300
S_2	P_4	400	S_1	P_4	400
S_3	P_1	200	S_1	P_1	200
S_3	P_2	200	S_1	P_2	200
S_3	P_3	300	S_1	P_3	300
S_3	P_4	400	S_1	P_4	400



Topic : SQL

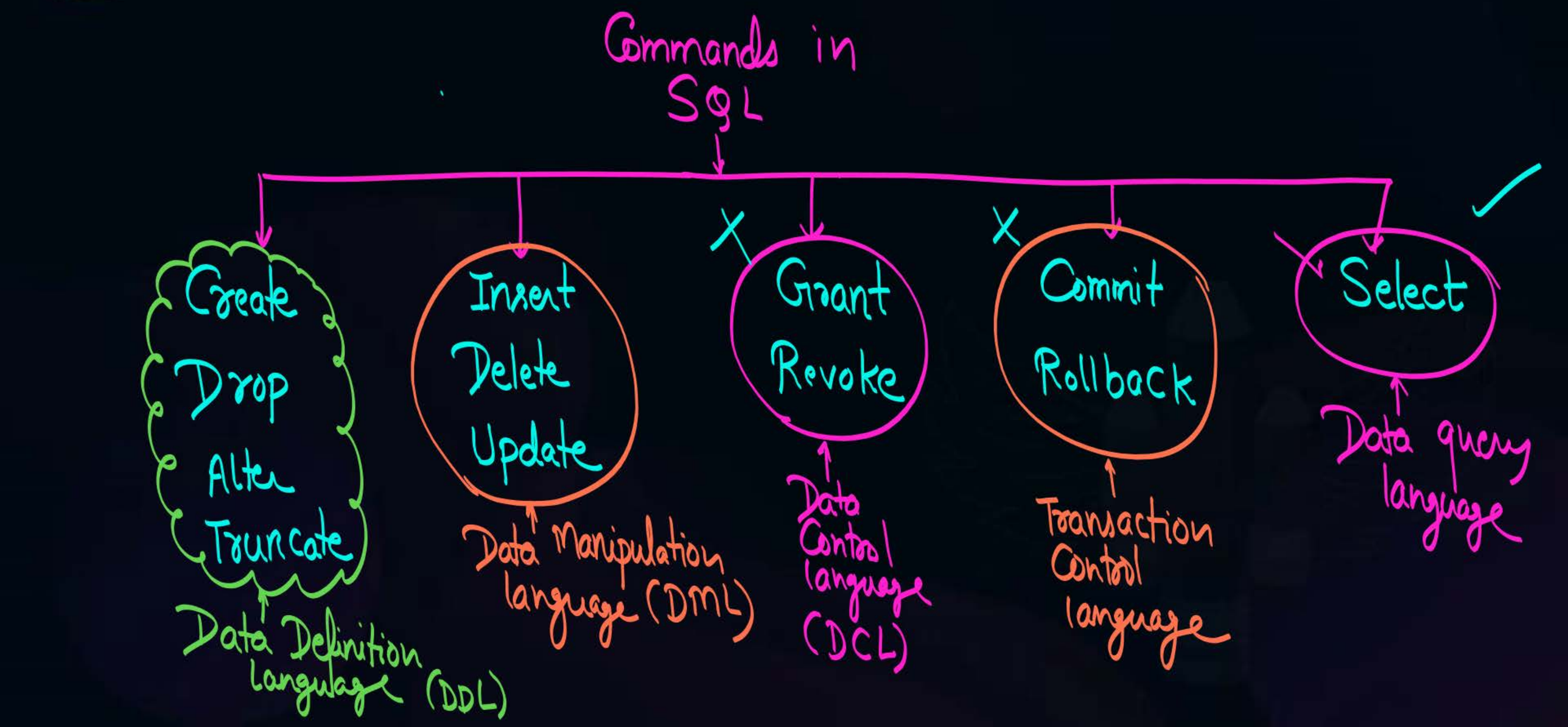
(Structured Query language)

It is a non-procedural query language



Topic : Commands in SQL







Topic : SQL clauses

Select [distinct]

distinct keyword is optional.

- If distinct keyword is used then there will be no duplicate tuples in the o/p
- without distinct keyword SQL query may produce duplicate tuples.

From :→

Where :→

Groupby :→

Having :→

Orderby :→



Topic : SQL clauses

Select [distinct] $A_1, A_2, \ldots A_n$

(list of attributes required in the o/p)

From :→ From $R_1, R_2, \ldots R_m$

(Names of relations)

Where :→

Group by :→

Having :→

Order by :→



Topic : SQL clauses

$\pi_{A_1, A_2, \ldots A_n} \equiv$ Select [distinct] $A_1, A_2, \ldots A_n$ ← list of attributes required in the o/p

From :→ From $R_1, R_2, \ldots R_m$ $\equiv R_1 \times R_2 \times \ldots R_m$

Where :→ Where (Condition) $\equiv \sigma_{Condition}$

Groupby :→ Groupby (Attributes)

Having :→ Having (Condition)

Orderby :→ Order by (Attribute)



Topic : Aggregate functions

There are different aggregate functions available in SQL

1) Count
2) Sum
3) Avg
4) Min
5) Max



Topic : Aggregate functions

1) Count (*) : It will return the total number of tuples in the relation

Count (Attribute): It will count the non-null values in the column corresponding to the attribute specified with Count aggregate function.

NULL + 5 = NULL

NULL :

1. Null is a non-zero unknown value
2. No two NULL are equal (NULL = NULL is not defined)
3. NULL values are always discarded by aggregate function.
4. Arithmatic opⁿ with NULL results NULL.
   eg. NULL + 10 = NULL



Topic : Aggregate functions

2) Sum (Attribute) : It will return the Summation of all non-null values w.r.t. Specified attribute.

Sum Distinct (Attribute): It will return the Summation of all distinct non-null values w.r.t. Specified attribute



Topic : Aggregate functions

3) Avg (Attribute) : It will return the average value w.r.t specified attribute

$$\text{Avg (Attribute)} = \frac{\text{Sum (Attribute)}}{\text{Count (Attribute)}}$$

+ $$\text{Avg Distinct (Attribute)} = \frac{\text{Sum Distinct (Attribute)}}{\text{Count Distinct (Attribute)}}$$



Topic : Aggregate functions

4) Min (Attribute) : It will return the minimum value from specified Column

5) Max (Attribute) : It will return the maximum value from specified Column.

* Count (*) from Student = 6

* Count (Marks) from Student = 5

* Count Distinct (Marks) from Student = 4

Student

Sid	Sname	Marks	Branch
S_1	A	40	CS
S_2	A	60	IT
S_3	B	20	Cs
S_4	C	60	EC
S_5	D	70	EC
S_6	C	NUL	IT

* Sum (marks) from Student = 250

* Sum Distinct (marks) from Student = 190

Student

Sid	Sname	Marks	Branch
S_1	A	40	CS
S_2	A	60	IT
S_3	B	20	CS
S_4	C	60	EC
S_5	D	70	EC
S_6	C	NUL	IT

* Avg (Marks) from Student $= \frac{250}{5} = 50$

* Avg Distinct (Marks) from Student $= \frac{190}{4}$

$= 47$

Student

Sid	Sname	Marks	Branch
S_1	A	40	CS
S_2	A	60	IT
S_3	B	20	Cs
S_4	C	60	EC
S_5	D	70	EC
S_6	C	NUL	IT

* Min(Marks) = 20

Max(Marks) = 70

Student

Sid	Sname	Marks	Branch
S_1	A	40	CS
S_2	A	60	IT
S_3	B	20	Cs
S_4	C	60	EC
S_5	D	70	EC
S_6	C	NUL	IT



2 mins Summary

- Topic: SQL
- Topic: SQL commands
- Topic: SQL clauses
- Topic: Aggregate functions

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THANK - YOU

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